



CEPIITD

Online PG Diploma: Advanced Communication Engineering with Quantum and AI Integration

Course: Quantum Information and Security

Instructor: Dr. Neel Kanth Kundu

Assistant Professor

Centre for Applied Research in Electronics (CARE)
Bharti School of Telecom Tech. & Mgmt. (Joint Faculty)
Indian Institute of Technology Delhi

Course Contents



- Classical Information Theory
 - Shannon's Source and Channel Coding Theorems
- Quantum Information Theory
 - Holevo Capacity
- Quantum Key Distribution
 - DV-QKD and CV-QKD Secrecy Analysis



Quantum Information and Security

Module-1

Classical Information Theory

Instructor: Dr. Neel Kanth Kundu

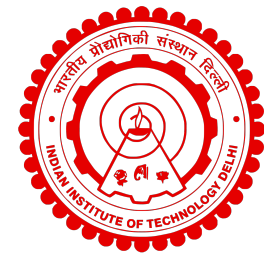
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Overview

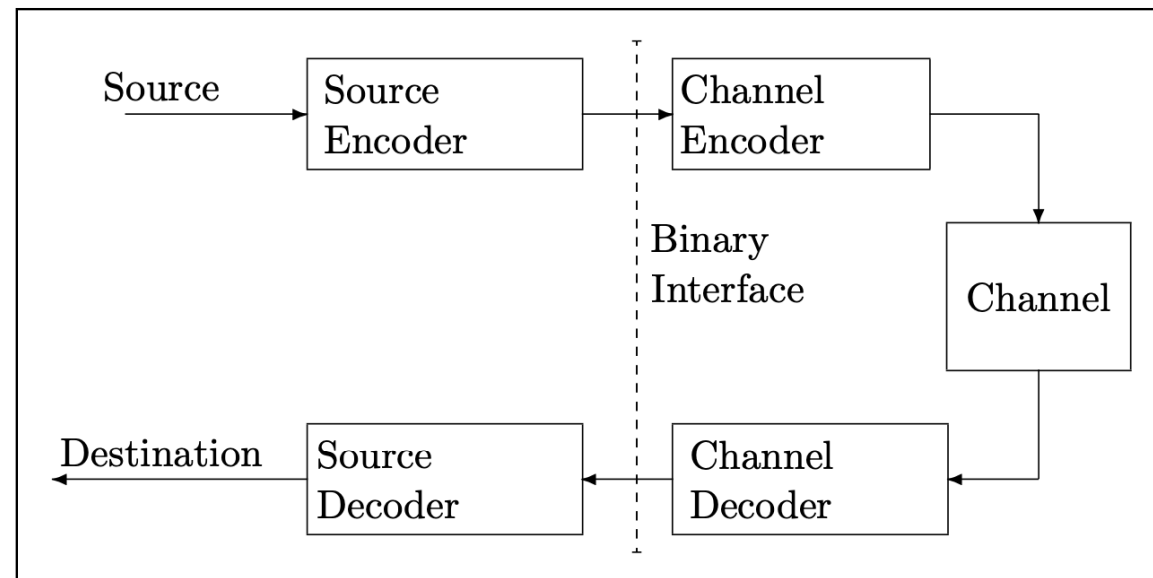


- Motivation
- Classical Entropy
- Noiseless Source Coding: Data Compression
- Noisy Channel Coding: Channel Capacity

Motivation



- Goals of Information theory :
 1. Define *information* mathematically and quantitatively
 2. Represent information in efficient way (data compression) for storage and transmission
 3. Ensure the protection of information (through encoding) in presence of noise & impairments
- Classical Information Theory (CIT): Achieve above goals in accordance with laws of classical physics
- Quantum Information Theory (QIT): Achieve above goals in accordance with quantum mechanical principles
- QIT is richer than CIT due to intriguing resources like entanglement



Classical Information Theory



- **Entropy** quantifies information and uncertainty.
- Important for **security proofs** in Quantum Key Distribution (QKD).
- Helps estimate adversary's knowledge in QKD.
- Understanding classical entropy builds intuition for quantum cases.
- **Claude Shannon** (1948) laid foundations of classical information theory.
- **Key Questions:**
 1. How many bits are needed to compress data reliably?
 2. How much information can be reliably transmitted through a channel?

Overview

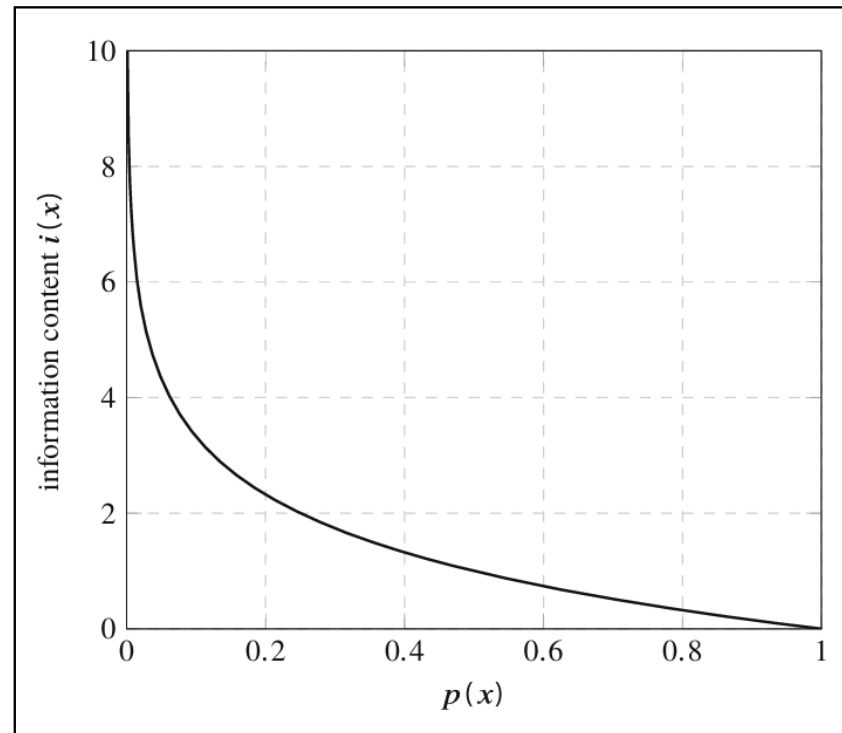


- Introduction
- **Classical Entropy**
- Noiseless Source Coding: Data Compression
- Noisy Channel Coding: Channel Capacity

Quantifying Information



- Let X be a random variable with realizations $x \in \mathcal{X}$ and probability $p_X(x)$
- Information content (surprise) of outcome x : $i(x) = -\log_2(p_X(x))$
- Units of information: **bits**



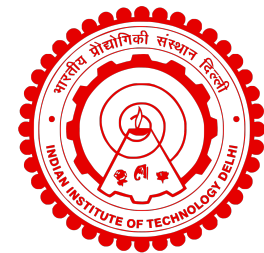


Properties of Information Content

- **Label-Invariance:**
 - Information content depends only on the probability of the outcome, not its label.
 - Example: Variables with values $\{0,1\}$ or $\{+, -\}$ and same probabilities carry the same information.
- **Continuity:** The function $i(x)$ is continuous for all $p_X(x) \in (0,1]$.
- **Monotonicity and Non-negativity:**
 - Unlikely events (small p) produce higher information content.
 - Since $\log_2(p) \leq 0$, the information content $i(x) \geq 0$ for all x .
- **Additivity for Independent Events:** If two random variables X_1 and X_2 are independent, then the total information content satisfies: $i(x_1, x_2) = i(x_1) + i(x_2)$
- When outcomes are independent, the total information is simply the sum of individual surprises.
- **Proof:**

$$\begin{aligned} i(x_1, x_2) &= -\log_2(p(x_1, x_2)) \\ &= -\log_2(p(x_1) \cdot p(x_2)) \quad (\text{independence}) \\ &= -\log_2(p(x_1)) - \log_2(p(x_2)) \\ &= i(x_1) + i(x_2) \end{aligned}$$

Shannon's First Question: Compression



- Q1: How many bits are required to reliably compress a given amount of information?
- A1: According to **Shannon's Noiseless Coding Theorem**, the optimal average number of bits required to encode symbols from a source with random variable X is given by its entropy.
- **Entropy:** $H(X) = - \sum_x p(x) \log p(x)$
- **Explanation:**
 - Entropy $H(X)$ measures the average uncertainty in a random variable.
 - It gives the lower bound for lossless compression — no method can, on average, compress below this limit without losing information.
 - For large enough sequences, it's possible to compress to approximately $H(X)$ bits per symbol.



Shannon Entropy

- **Goal:** Measure the average information content of a random variable X .
- **Definition:** The entropy of a discrete random variable X with probability distribution $p_X(x)$ is:
$$H(X) = - \sum_x p_X(x) \log_2 p_X(x)$$
 - Convention: $0 \cdot \log(0) = 0$
 - Justification: $\lim_{p \rightarrow 0} p \log p = 0$
- **Example-1:** Fair Coin Toss
 - Let X be the outcome of a fair coin: $p(H) = p(T) = \frac{1}{2}$.
 - Then $H(X) = - \left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) = 1$ bit
 - **Interpretation:** Each outcome gives us 1 bit of information.
- **Example-2:** A fair 8-sided die: $P(i) = \frac{1}{8}, i = 1, \dots, 8$
 - Entropy: $H(X) = - \sum_{i=1}^8 \frac{1}{8} \log_2 \left(\frac{1}{8} \right) = \log_2(8) = 3$ bits



Binary Entropy

- Let X be a random variable with values in $\{0,1\}$.
- Probability distribution:
 $P(X = 0) = p, P(X = 1) = 1 - p$
- Binary Entropy is defined as:
 $h_2(p) = -p \log p - (1 - p) \log(1 - p)$
- Symmetry: $h_2(p) = h_2(1 - p)$
- Maximum entropy at $p = 0.5$ (unbiased coin)
- Minimum entropy ($h_2(p) = 0$) when $p = 0$ or 1 (deterministic)

